

Kaiwen Lin

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🐙 GitHub Profile

EDUCATION

- **The University of Texas at Austin** Aug 2024 – May 2026
Master of Science in Information Science GPA: 3.91/4.0;
 - **Courses:** Algorithm; Multicore Computing/ Parallel Computing; Data Warehousing; Deep Learning and Multimodal Systems; Database Management; Machine Learning; JavaScript
- **University of Chinese Academy of Social Sciences** 2020 - 2023
M.A. in International Political Economy Graduate with Distinction, GPA: 3.9/4.0;

EXPERIENCE

- **Austin AI Hub** Jan 2026 – May 2026
AI Engineer Intern Austin, TX
 - Orchestrated a multi-agent system using LangGraph, implementing stateful cyclic graphs and Hybrid RAG, architecting an extensible Model Context Protocol middleware layer to standardize tool invocation and context injection across heterogeneous agents; leveraged LangSmith for granular trace observability and automated evaluation datasets to enforce enterprise-grade AI safety and quality gates
 - deployed a full-stack architecture, building a responsive React frontend and FastAPI backend to enable real-time streaming of multi-agent reasoning traces.
- **The Institute of World Economics and Politics, Chinese Academy of Social Sciences** Nov 2020– May 2023
Research Assistant (Data Platform / Data Science) Beijing
 - Built and maintained Airflow ETL pipelines for AWS S3 data ingestion and migrated multi-billion-row econometric datasets to Snowflake, optimizing schemas to reduce query latency by 90% and prep time by 50%.

PROJECT

- **Email Agent: Prompt Injection & Reinforcement Learning (GRPO) Defense** May 2026 - present
LangGraph, PAIR, SFT, GRPO, QLoRA, ModernBERT, HuggingFace TRL, PyTorch
 - Built a 5-tool LangGraph ReAct email agent and an automated red-teaming pipeline; implemented a PAIR-style adversarial loop with an LLM judge to generate indirect prompt injection rollouts across three attack vectors (instruction override, hidden injection, data exfiltration).
 - Developed a dual-layer defense architecture: trained a ModernBERT-base classifier to gate destructive tool calls and aligned a Qwen2.5-1.5B model using SFT and GRPO (Group Relative Policy Optimization); diagnosed TRL gradient checkpointing bugs and analyzed reward hacking behaviors to optimize the Attack Success Rate vs. benign pass rate trade-off.
- **GPU-Accelerated LU Factorization (CUDA)** March 2026 - April 2026
CUDA, C++, Shared Memory Tiling, Binary Search, CUDA Events
 - Investigated when sparse GPU LU factorization becomes faster than dense LU by implementing custom dense untiled, dense tiled, and sparse CSC factorization kernels and benchmarking the sparsity crossover across matrix sizes.
 - Built a correctness-checked benchmark pipeline and showed that larger matrices shift the dense-sparse threshold to lower sparsity levels.
- **Distributed Message Queue in Go** Nov 2024 – Mar 2025
Go, Custom TCP Binary Protocol, Write-Ahead Log, Raft Leader Election, Prometheus
 - Built a topic-based pub/sub broker with at-least-once delivery via segmented WAL and consumer ACK tracking; deployed a 3-node cluster with simplified Raft leader election and configurable async/sync replication.
 - Designed an RDY-based flow-control protocol over raw TCP to back-pressure consumers; optimized throughput from 2.5K to 8K msgs/sec by replacing mutex-guarded publishes with channel-based single-writer pipelines and 50ms batched `fdasync`.

SKILLS

Programming Languages: Java, Python, SQL, Go, C++

Data Infrastructure: dbt, Spark, Kafka, Vector Databases, Hadoop

Machine Learning & AI: PyTorch, RAG, LLMs, Reinforcement Learning (GRPO, OPD), CUDA

Cloud & DevOps: GCP (Bigquery, Vertex AI), AWS (S3), Snowflake, Docker, Kubernetes, CI/CD, Git

Languages: Mandarin Chinese (native); English (fluent)